

Use case	Problem without observability	How open source solves it
RAG pipeline quality	Retrieval quality degrades silently as data changes	Trace retrieval latency, chunk relevance scores, and context token usage using OpenLLMetry + Arize Phoenix
Agentic workflow debugging	Multi-step agents fail non-obviously — wrong tool selection, infinite loops, silent errors	Full span traces per agent run (Jaeger/Tempo), tool call success/failure metrics, loop detection via trace depth alerts
Multi-model cost governance	Multiple LLMs across teams create uncontrolled spend with no attribution	Token consumption dashboards (Grafana + Prometheus), cost-per-workflow alerting, model routing efficiency via OpenCost
Hallucination and drift detection	Output quality degrades over time as prompts age or models update	Evidently AI for output drift detection; LangFuse for eval scoring; alert when quality metrics fall below SLO thresholds
LLM SLA enforcement	SLAs are now expected for AI features — requires observability infrastructure	Prometheus + Alertmanager for p95 latency SLOs; Grafana dashboards for real-time SLA burn rate tracking
Security and compliance auditing	AI systems subject to audit — prompt logs and output histories must be captured and governed	Structured logging via Loki with access controls; retention policies aligned to compliance requirements (GDPR, HIPAA)
Fine-tuning and experiment tracking	Without tracking, fine-tuning runs are unreproducible — no ability to correlate model version to prod behaviour	MLflow for experiment logging; link model registry entries to production observability dashboards

Observability is the discipline that makes AI trustworthy

We are at an inflection point. The first wave of enterprise AI was about proving it works. The second wave — the one most organisations are entering now — is about making it reliable, governable, and trustworthy enough to run in mission-critical contexts. That transition cannot happen without observability.

The shift from black box to glass box is not primarily a tooling challenge — the open source ecosystem has solved that. It is an architectural commitment: to instrument pipelines before they go to production, to define quality SLOs alongside latency SLOs, to build feedback loops from production telemetry into evaluation pipelines, and to treat prompts as versioned artifacts with observable lifecycles.

Organisations that build this capability now will have a structural advantage. They will catch degradation before users do. They will govern costs before they spiral. They will debug in hours rather than days. And they will have audit trails, quality histories, and model performance data that regulators and auditors are increasingly requiring.

The most important architectural decision you can make for your AI investments in 2026 is not which model to run. It is whether you can see what it is doing once it is running.  

References

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